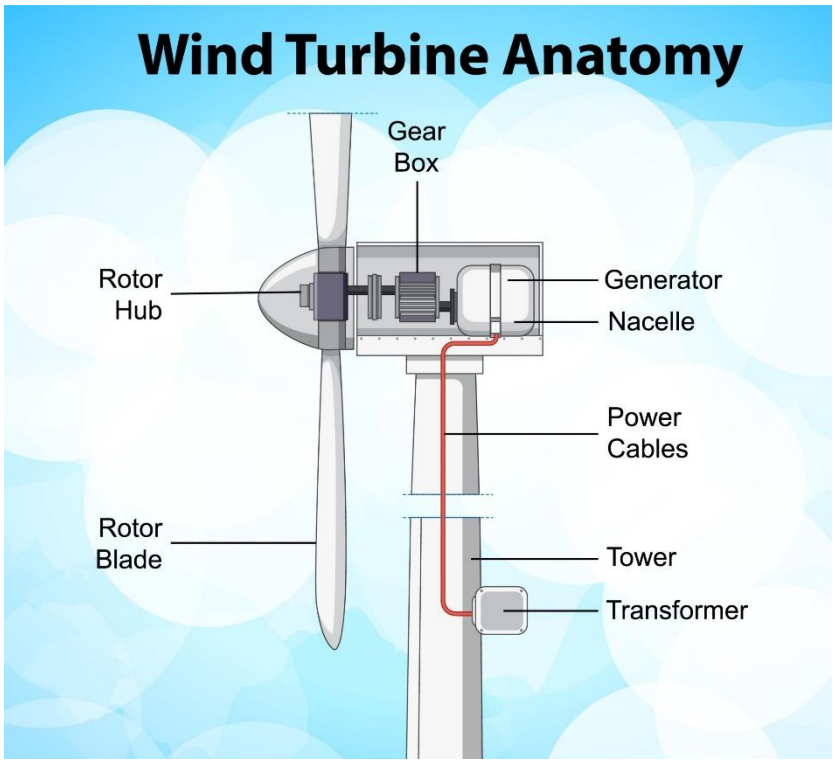
UNIT 4: WIND ENERGY TECHNOLOGY

1. Atmospheric Origin & Distribution of Winds

Origin of Winds

- **Brief Description:** Wind is the large-scale movement of air relative to the Earth's surface, driven primarily by uneven solar heating of the planet and the Earth's rotation.
- **Long Description:**
    - **Solar Heating & Pressure Gradients:** Solar radiation heats equatorial regions more intensely than polar regions. Warm air expands, becomes lighter, and rises, creating low-pressure areas near the surface. At the poles, cold air becomes dense, sinks, and creates high-pressure areas. Air flows naturally from high-pressure zones to low-pressure zones, creating a **Pressure Gradient Force (PGF)**.
    - **Global (Planetary) Winds:** As air moves, it is deflected by the Coriolis effect (caused by Earth's rotation to the east). In the Northern Hemisphere, winds deflect to the right; in the Southern Hemisphere, to the left. This creates global circulation cells (Hadley, Ferrel, and Polar cells) and planetary wind belts: Trade Winds, Westerlies, and Polar Easterlies.
    - **Local Winds:** Driven by localized temperature and pressure differences caused by topography, land-sea thermal contrasts (sea breezes and land breezes), and mountain-valley thermal cycles (katabatic and anabatic winds).



Factors Affecting Distribution of Wind Energy on Earth's Surface

- **Brief Description:** Geographical location, topography, surface roughness, altitude, and seasonal climate cycles dictate the spatial and temporal distribution of wind energy.
- **Long Description:**
    - **Latitude & Atmospheric Circulation:** Regions under persistent high or low-pressure belts (like the doldrums vs. the roaring forties) experience vastly different average wind speeds.
    - **Topography & Orographic Effects:** Valleys, mountain passes, and coastal headlands funnel or accelerate airflow through Venturi effects, creating high-wind corridors.
    - **Surface Roughness & Obstacles:** Forests, cities, and hills create friction that slows down surface winds, creating a turbulent boundary layer. Oceans, flat plains, and ice sheets offer minimal resistance, allowing high wind energy potential.
    - **Diurnal and Seasonal Variations:** Solar heating cycles cause daytime thermal mixing (stronger winds) and nighttime stratification (weaker winds), while seasonal shifts alter jet stream positions.

2. Wind Measurement, Statistics, and Site Estimation

Wind Speed and Direction Measuring Instruments

- **Brief Description:** Meteorological instruments quantify wind behavior for resource assessment.
- **Long Description:**
    - **Anemometers (Wind Speed):**
        - *Cup Anemometer:* Three or four hemispherical cups rotate around a vertical axis; rotation speed is proportional to wind speed.
        - *Sonic Anemometer:* Measures the transit time of ultrasonic sound pulses between transducers to calculate 3D wind velocity vector components without moving parts.
        - *Propeller Anemometer:* Uses a small propeller aligned with the wind via a tail fin.
    - **Wind Vanes (Wind Direction):** A asymmetric fin or airfoil that points into the wind, coupled with a potentiometer or digital encoder to output angular direction (0° to 360°).

- **Remote Sensing (SODAR & LiDAR):** Sound Detection and Ranging (SODAR) and Light Detection and Ranging (LiDAR) transmit acoustic or laser pulses into the atmosphere and measure Doppler shifts in backscattered signals to profile wind speeds at hub heights up to several hundred meters.

Variation of Wind Speed with Height

- **Brief Description:** Wind speeds increase with height above ground due to a reduction in surface friction.
- **Long Description:** The vertical wind profile within the atmospheric boundary layer is modeled using empirical laws:
  - **Logarithmic Wind Profile:** Based on boundary layer fluid mechanics:

$$v(z) = \frac{u_*}{\kappa} \ln\left(\frac{z}{z_0}\right)$$

where $v(z)$ is speed at height z, u_* is friction velocity, κ is von Kármán constant (~0.4), and z_0 is surface roughness length.

  - **Power Law Profile (Hellman Exponential Law):** Widely used in engineering practice:

$$\frac{v}{v_{ref}} = \left(\frac{z}{z_{ref}}\right)^{\alpha}$$

where α is the Hellman exponent (friction coefficient), ranging from ~0.10 for smooth open water to ~0.40 for dense urban environments or heavy forests.

Estimation of Wind Energy at Site & Weibull Distribution

- **Brief Description:** Evaluating site feasibility requires statistical analysis of wind speed distributions and available power density.
- **Long Description:**
  - **Wind Power Density:** The kinetic power available per unit swept area (P/A) is given by:

$$\frac{P}{A} = \frac{1}{2}\rho v^3$$

where ρ is air density (≈ 1.225 kg/m^3 at sea level) and v is wind speed. Because power varies with the cube of wind speed, accurate wind speed distribution is critical.

- **Weibull Probability Distribution Function (PDF):** Wind speeds fluctuate randomly; their frequency distribution is best modeled by the two-parameter Weibull distribution:

$$f(v) = \left(\frac{k}{c}\right)\left(\frac{v}{c}\right)^{k-1} \exp\left[-\left(\frac{v}{c}\right)^{k}\right]$$

  - k = **Shape factor** (dimensionless): indicates the stability or peakedness of wind speeds (typically between 1.5 and 2.5).
  - c = **Scale factor** (m/s): closely related to the mean wind speed of the site.
- **Rayleigh Distribution:** A special case of the Weibull distribution where the shape factor $k = 2$, often used when only mean wind speed data is available.

3. Site Selection & Capacity Factor

Site Selection Criteria for Wind Energy

- **Brief Description:** Choosing a wind farm location balances high wind resource availability with economic, logistical, and environmental constraints.
- **Long Description:**
  - **High Wind Resource:** Annual average wind speeds generally must exceed 6–7 m/s at hub height for utility-scale commercial viability.
  - **Terrain & Accessibility:** Flat or gently rolling terrain, ridgelines, or shallow offshore waters with good port access and road infrastructure for heavy crane transport.
  - **Grid Proximity:** Short interconnection distance to high-voltage transmission lines to minimize electrical transmission losses and capital costs.
  - **Environmental & Social Constraints:** Avoidance of protected ecological zones, bird migration corridors, military airspace, residential noise buffer zones, and cultural heritage sites.

Capacity Factor

- **Brief Description:** The ratio of actual energy produced by a wind turbine over a given period to the energy that would have been produced if it operated continuously at rated full power.

- **Long Description:**
    - Capacity Factor (CF) $= \frac{\text{Actual Annual Energy Output (MWh)}}{\text{Rated Power (MW)} \times 8760 \text{ hours}}$
    - Typical onshore wind turbines achieve CFs between 25% and 45%, while offshore installations often reach 40% to 60% due to steadier, stronger marine winds.

4. Turbine Aerodynamics & Power Extraction

Extraction of Wind Turbine Power

- **Brief Description:** Wind turbines convert the kinetic energy of moving air into mechanical shaft rotation and subsequently into electrical energy.
- **Long Description:**
    - **Wind Power Formula:**

$$P_{wind} = \frac{1}{2}\rho A v^3$$

    - **Betz Model & Betz Limit:** Albert Betz (1919) used momentum theory applied to an idealized actuator disc to prove that no wind turbine can capture more than a theoretical maximum of **56.3%** $(16/27)$ of the kinetic energy in the wind.

$$P_{max} = \frac{16}{27}\left(\frac{1}{2}\rho A v^3\right) \approx 0.593 P_{wind}$$

The power coefficient C_p represents efficiency: $C_p = P_{turbine}/P_{wind}$, where $C_{p,max} = 0.593$ (Betz Limit). Modern high-performance turbines achieve practical C_p values between 0.45 and 0.50.

Aerodynamic Operation: Drag vs. Lift

- **Brief Description:** Wind turbine blades harness aerodynamic forces—primarily lift—to rotate the rotor.
- **Long Description:**
    - **Drag-based Turbines:** Rely on the direct push of wind against a surface (e.g., Savonius rotors). They have low rotational speeds and lower efficiency ($C_p \approx 0.15$).
    - **Lift-based Turbines:** Modern blades act like airplane wings (airfoils). As air flows over the curved blade profile, a pressure difference is created, generating an aerodynamic **lift force** perpendicular to the relative wind direction, along with a smaller **drag force** parallel to it. Lift forces create much higher torque and rotational velocities, enabling high efficiency.

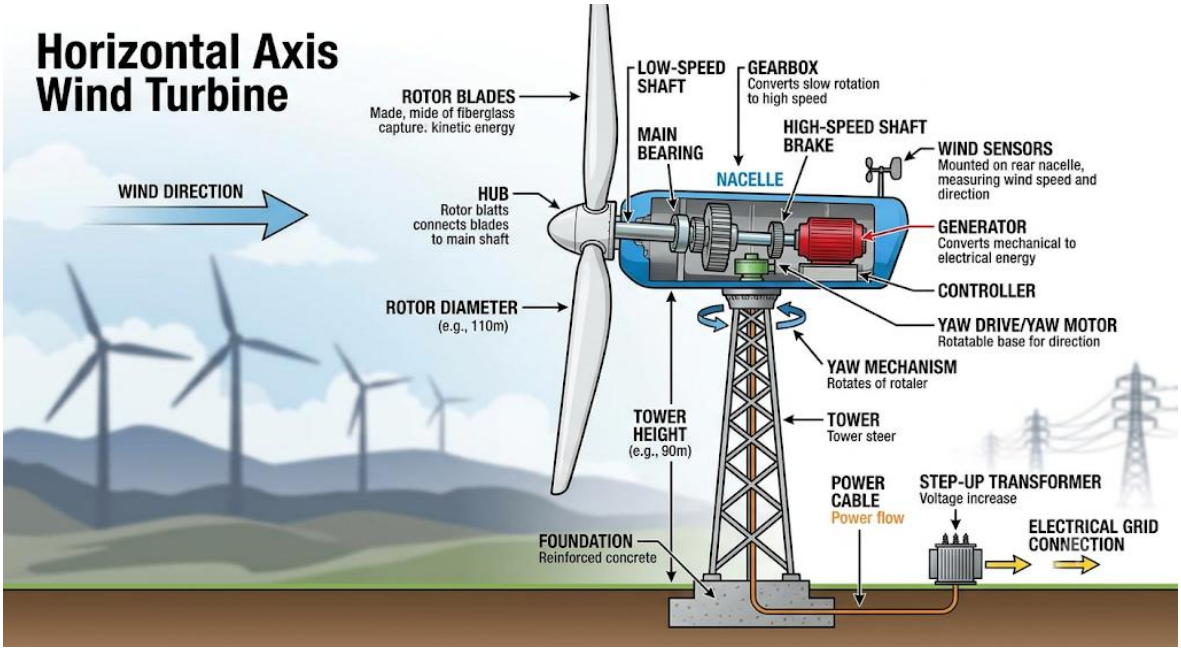
5. Turbine Classification, Rotor Types, & Components

Types of Wind Turbines

- **Horizontal Axis Wind Turbines (HAWT):** The axis of rotor rotation is parallel to the wind stream. Requires a yaw mechanism to face the wind. Dominates utility-scale power generation.
- **Vertical Axis Wind Turbines (VAWT):** The axis of rotation is perpendicular to the ground and wind stream (e.g., Darrieus and Savonius designs). Omnidirectional (accepts wind from any direction without a yaw system), with generator and gearbox mountable at ground level, though generally less efficient and subject to cyclic fatigue stresses.

Main Components of a HAWT

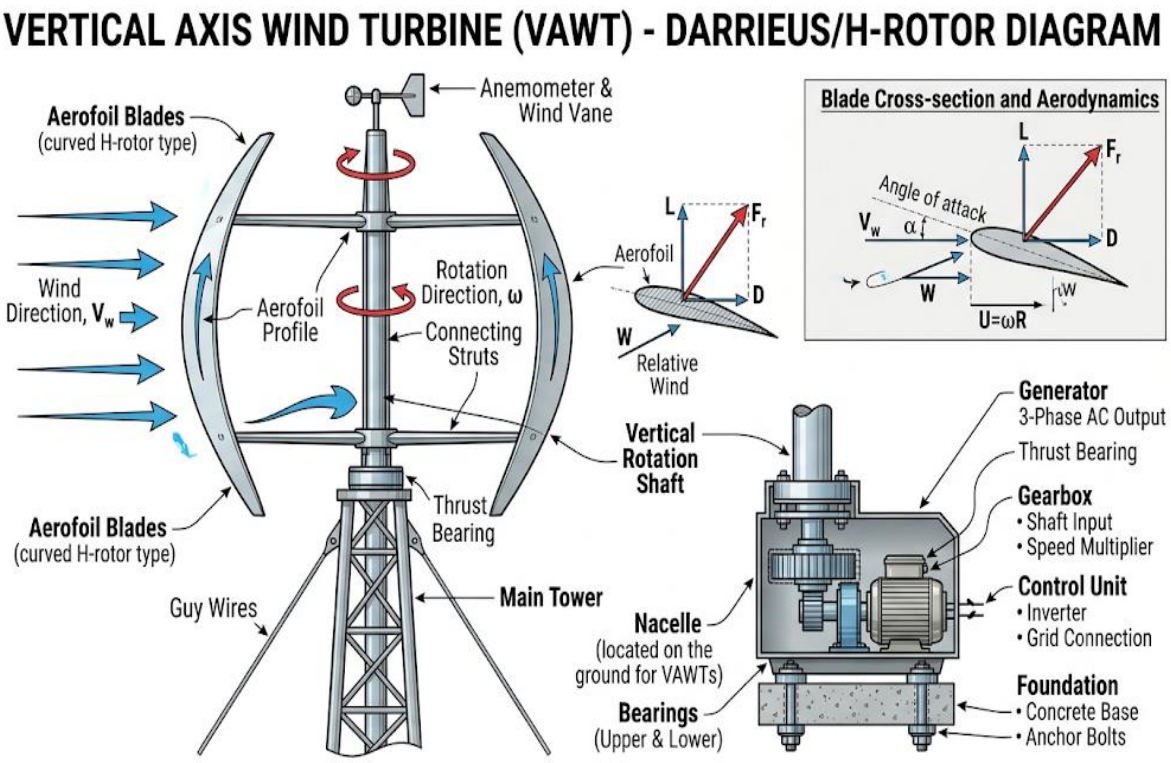
- **Rotor Blades:** Aerodynamic airfoils (typically 3 blades made of fiberglass-reinforced polyester or epoxy) that capture wind energy.
- **Hub:** Central structure connecting the blades to the main shaft.
- **Nacelle:** Enclosure housing internal machinery perched atop the tower.
    - *Low-speed shaft:* Connects rotor hub to gearbox.
    - *Gearbox:* Increases rotational speed (typically from ~10–20 rpm to 1500–1800 rpm for standard generators). Gearless (direct-drive) designs eliminate this via multi-pole synchronous generators.
    - *Generator:* Converts mechanical energy into electricity (Doubly Fed Induction Generators or Permanent Magnet Synchronous Generators).
    - *Yaw system:* Motors and gears that rotate the nacelle to align with wind direction.
    - *Brake system:* Mechanical disc brakes for emergency stopping or rotor locking during maintenance.
- **Tower:** Tubular steel or concrete structure supporting the nacelle at optimal height (typically 80–150+ meters).
- **Foundation:** Massive reinforced concrete base anchoring the tower to the ground.



Main Components of a VAWT

- **Rotor Blades**
- Function: Capture the kinetic energy of the wind and convert it into rotational torque around the main axis.
- Types:
    - *Darrieus (Lift-driven):* Aerofoil-shaped blades (straight H-rotor or curved troposkien shape) that operate using aerodynamic lift at higher tip-speed ratios.
    - *Savonius (Drag-driven):* S-shaped or scoop-like blades driven primarily by aerodynamic drag, offering high starting torque at low wind speeds.
- **Rotor Shaft (Central Shaft / Upright Stanchion)**
- Function: Serves as the central vertical axis of rotation. Transmits the mechanical torque generated by the rotating blades down to the drive train components.
- **Hub & Connecting Struts (Link Rods)**
- Function: Mechanically join the rotor blades to the main vertical shaft. Struts provide structural rigidity against centrifugal forces during high-speed rotation.

- **Gearbox (Optional / Drive Train)**
- Function: Steps up the low rotational speed of the rotor shaft to match the higher operational speeds required by standard electrical generators. Direct-drive configurations bypass the gearbox to reduce mechanical losses.
- **Generator**
- Function: Converts the mechanical shaft power into electrical power (AC output).
- Advantage in VAWTs: Located at or near ground level inside a base enclosure, making maintenance significantly easier and reducing top-heavy structural loading compared to HAWT nacelles.
- **Braking System**
- Function: Employs mechanical or electromagnetic resistance to slow down or lock the rotor during extreme wind events (cut-out speeds) or routine maintenance.
- **Support Tower & Base Foundation**
- Function: Elevates the rotor assembly and anchors the entire system to the ground, absorbing static, dynamic, and bending loads. Ground-level mounting reduces the required structural thickness of the tower foundation compared to equivalent HAWT designs.



6. Wind Energy Conversion Systems (WECS) & Storage

System Configurations & Modes of Operation

- **General Block Diagram of WECS:**

  Wind → Rotor Blades → Drive Train (Gearbox/Shaft) → Generator
  → Power Electronic Converters → Transformer → Grid / Load

- **Stand-Alone (Off-Grid) Mode:** Operates independently in remote locations, often paired with battery banks, solar PV, and diesel backup generators to ensure uninterrupted power supply.
- **Backup Mode (Wind-Diesel):** Combines wind turbines with diesel generator sets. A control system optimizes diesel operation, shutting down or idling diesel engines when wind penetration is high.
- **Grid-Connected Mode:** Integrates wind farms directly into regional power transmission grids. Requires advanced power electronics (inverters) to synchronize frequency, voltage, and phase angle.

Types of Generators Used in WECS

- **Squirrel-Cage Induction Generator (SCIG):** Robust and simple, but operates at a fixed speed (constant slip).
- **Doubly-Fed Induction Generator (DFIG):** Allows variable-speed operation through partial-scale frequency converters connected to the rotor winding, improving capture efficiency.
- **Permanent Magnet Synchronous Generator (PMSG):** Used in gearless direct-drive systems; high efficiency, reliable, but requires full-scale power electronic converters.

Wind Energy Storage

- **Brief Description:** Mitigates wind intermittency by storing surplus energy during high wind periods and discharging it during lulls.
- **Long Description:** Technologies include Lithium-ion and Redox Flow battery energy storage systems (BESS), Pumped Hydro Storage (PHS), Compressed Air Energy Storage (CAES), and Power-to-X (green hydrogen production via electrolysis).

7. Environmental Aspects, Pros, Cons, & Applications

Environmental Aspects

- **Positive Impacts:** Displaces fossil fuel combustion, significantly reducing greenhouse gas emissions (CO_2, SO_x, NO_x) and water consumption.

- **Negative Impacts & Mitigation:**
    - *Avian & Bat Collisions:* Mitigated by pre-construction siting studies, radar-based curtailment, and acoustic deterrents.
    - *Noise Emissions:* Aerodynamic and mechanical noise managed through turbine setback distances and modern blade design.
    - *Visual & Land Use:* Addressed via careful landscape integration; land underneath turbines remains available for agriculture or grazing.

Advantages & Disadvantages Summary

Advantages	**Disadvantages**
Clean, inexhaustible, renewable energy source	Intermittent and variable resource availability
Low operational and maintenance costs post-installation	High initial capital investment
Dual land use (farming/grazing coexist with turbines)	Potential aesthetic and visual impact
Rapid deployment capability (onshore and offshore)	Grid congestion and transmission infrastructure requirements

Major Applications of Wind Power

1. **Utility-Scale Electricity Generation:** Onshore and offshore wind farms feeding regional transmission grids.
2. **Distributed & Community Wind:** Medium- and small-scale turbines powering remote communities, islands, farms, and industrial facilities.
3. **Off-Grid Telecommunications & Pumping:** Powering remote cell towers, navigational buoys, and agricultural water pumping systems.